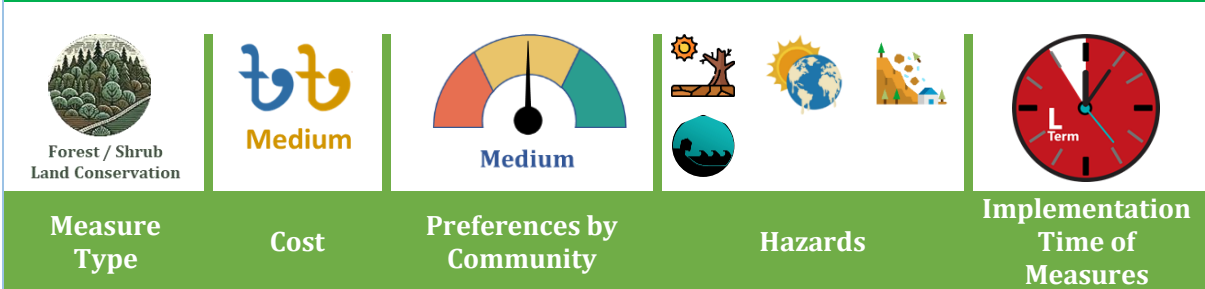


## F-7. Forest Fire Management



## Descriptions

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |           |      |            |      |            |      |           |  |
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| Measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Forest fire management refers to the strategic actions and practices aimed at preventing, controlling, and mitigating the effects of wildfires in forested areas. It involves a combination of proactive measures to reduce the risk of forest fires, as well as responsive strategies to manage and extinguish fires once they occur. Effective forest fire management is essential to protect forests, wildlife, communities, and natural resources.                                                                                                                                                                                                                                                                                    |      |           |      |            |      |            |      |           |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | The CHT' diverse biodiversity, steep topography, and dependence on traditional farming methods create particular difficulties for managing forest fires. Particularly during the dry season, the region is vulnerable to forest fires, which can have catastrophic consequences for the local population and ecosystem. However, forest fire risks and impacts can be greatly decreased with a coordinated strategy that includes prevention, early detection, firefighting preparation, and post-fire recovery. The region's fire management system will become more resilient and successful with the involvement of local communities, the integration of indigenous knowledge, and the maintenance of sustainable land use practices. |      |           |      |            |      |            |      |           |  |
| Location                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | R-114                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | R-99 | R-65      | K-98 | K-76       | K-71 | B-173      | B-77 | B-76      |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | ■                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | ■    | ■         | ■    | ■          | ■    | ■          | ■    | ■         |  |
| Applicable Sites                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Suitable for degraded lands, barren area, forest                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |           |      |            |      |            |      |           |  |
| Slope                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Less than 3%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      | 3% to 15% |      | 15% to 30% |      | 30% to 60% |      | Above 60% |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |           |      |            |      |            |      |           |  |
| Design options and performance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |           |      |            |      |            |      |           |  |
| Feasibility criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |           |      |            |      |            |      |           |  |
| <ul style="list-style-type: none"><li>In the CHT, where fires are more likely to occur during the dry months, fire management plans should take into consideration the monsoon and dry seasons. Prioritization must also be given to soil conservation and erosion control strategies, such as restoring vegetation after fires. To prevent harming sensitive species or their ecosystems, methods like controlled burns should be modified to suit local conditions.</li><li>Adjusting strategies in real time requires a system for tracking fire risk, identifying fires early, and evaluating the efficacy of management techniques. Successful fire management requires the active involvement of local people, especially indigenous populations. Their expertise, experience, and active participation in firefighting and fire prevention initiatives are essential.</li></ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |           |      |            |      |            |      |           |  |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |               |             |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------|-------------|
|                                  | <ul style="list-style-type: none"> <li>Long-term sustainability in fire management systems should provide sustained financial support, community involvement, and the flexibility of fire management techniques throughout time. Future environmental changes, such as an increase in the frequency or intensity of wildfires driven on by climate change, should be addressed by strategies that are adaptable enough.</li> </ul>                                                   |            |               |             |
| <b>Operation and maintenance</b> | It is crucial to regularly monitor and maintain high-risk locations with surveillance equipment like drones, satellite imagery, and watchtowers. Teams fighting fires need to be properly educated and outfitted with the right equipment. In order to decrease fires caused by humans, ongoing community education and awareness campaigns are also crucial. In order to recover ecosystems following fires, the area has to concentrate on reforestation and soil erosion control. |            |               |             |
| <b>Cost</b>                      | Medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |               |             |
| <b>Benefits</b>                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |               |             |
|                                  | <b>Type</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Low</b> | <b>Medium</b> | <b>High</b> |
|                                  | Canopy Cover Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |               |             |
|                                  | Carbon Sequestration                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |               |             |
|                                  | Crop Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |               |             |
|                                  | Crop Failure Risk Reduction                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |               |             |
|                                  | Ensuring Food Security                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |               |             |
|                                  | Fruit Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |               |             |
|                                  | Forestry Goods Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |               |             |
|                                  | Livestock Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |               |             |
|                                  | Fisheries Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |               |             |
|                                  | Alternative Livelihood Generation                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            |               |             |
|                                  | Increase of Income                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |               |             |
|                                  | Soil Erosion Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |               |             |
|                                  | Improved Soil Health                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |               |             |
|                                  | Soil Fertility Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |               |             |
|                                  | Sustainable Water Supply                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |               |             |
|                                  | Water Quality Improvement                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |               |             |
|                                  | Water Purification                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |               |             |
|                                  | River Bank Erosion Control                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |               |             |
|                                  | Surface Runoff Reduction                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |               |             |
|                                  | Groundwater Recharge                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |               |             |
|                                  | Pest and Disease Management                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |               |             |
|                                  | Reduction of Pesticides and Fertilizer Use                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |               |             |
|                                  | Ecosystem and Biodiversity Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                  |            |               |             |
|                                  | Better Communications                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |               |             |
|                                  | Better Education                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |               |             |
|                                  | Better WASH facilities                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |               |             |
| <b>Implementation Challenges</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |               |             |

|                                  | Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Low | Medium | High |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--------|------|
|                                  | Land unavailable/high land area required                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |        |      |
|                                  | Supply chain disconnection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |        |      |
|                                  | Land right related problems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |        |      |
|                                  | High human investment capital need                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |        |      |
|                                  | Insufficient infrastructures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |        |      |
|                                  | Less demand of agroforestry product                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |        |      |
|                                  | Environmentally unsustainable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |        |      |
|                                  | Socially unacceptable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |        |      |
|                                  | Less beneficial                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |        |      |
| <b>Environmental performance</b> | <ul style="list-style-type: none"> <li>Numerous plant and animal species in the area are guaranteed to survive thanks to forest fire management, which helps stop habitat damage. In addition to providing vital wildlife habitat, healthy forests additionally assist to preserve biodiversity.</li> <li>Ecosystems are encouraged to naturally regenerate by lessening the frequency and intensity of forest fires. This keeps forest ecosystems healthy and improves their capacity to bounce back from shocks.</li> <li>As carbon sinks, forests remove carbon dioxide from the atmosphere. The carbon deposited in trees and other plants is protected by preventing forest fires, which aids in the fight against climate change.</li> <li>Ash along with other dangerous pollutants, such as particulate matter, are released into the air and water by forest fires. Effective fire control reduces these emissions, enhancing air quality and lowering aquatic contamination.</li> </ul> |     |        |      |
| <b>Remarks</b>                   | This measure will aid landscape restoration in CHT of Bangladesh along with creating alternative livelihood opportunities.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |        |      |